Signals beyond VRP: what else the desk can trade on with options

Strategy research · L/S Gamma Desk, QUANTT

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Question: look into how we can incoorporate additional signals beyond VRP to make trades on, this can even go beyond a gamma scalping trade or long short gamma. as long as the trade is options related

Method: Searched all 35 papers in `docs/papers/` (page-tagged index) + web search. Quotes verified against the cited PDF pages. Page numbers are PDF pages; printed page numbers are given where they differ. Web sources were checked on their abstract pages (IDEAS/RePEc, NBER, journal or author pages). One desk data check on the registered `always25` backtest (aggregates only).

Previous research: Signal generation for market state · Volatility surface for the strategy · Shock or regime trigger for long gamma

1 Short answer

The best-supported signals beyond VRP fall into four groups:

1. **When to hold.** Calendar timing: weekends, overnight, FOMC days.
2. **How much to hold.** VVIX and the VIX term-structure slope.
3. **Which option to sell.** Skew and term structure, used to choose the strike and expiry rather than whether to trade.
4. **Where to sell.** Volatility premia on other ETFs.

Two less obvious points matter for the desk:

- **Calendar timing has the cleanest evidence, and our own data agrees.** Option returns are lower over weekends and overnight (Jones & Shemesh 2018; Muravyev & Ni 2020). In our `always25` backtest, the short put made essentially all of its mark-to-market profit between Friday 10:15 and Tuesday 10:15, in both halves of 2014-2024.
- **Several popular signals are weak or redundant:**
  - Skew trades are mostly the same bet as VRP (Kozhan et al. 2013).
  - Trend added nothing to short-put selection in the one SPX paper that tested it.
  - 0DTE rules lose money after correctly measured costs (Vilkov's 2026 correction).
  - Dealer-gamma signals need open-interest data the desk doesn't have historically.

The library itself has only one paper that tests signals for SPX option selling: the learning-to-rank paper. Most of the evidence below comes from the web.

2 What the papers say

2.1 What the library covers

- The library is mostly about VRP forecasting, regimes and hedging. Only Wysocki's learning-to-rank paper tests a broad signal set for SPX option selling.
  - It picks one of eight delta-targeted short puts, or no trade, each day.

    > "A LightGBM LambdaRank ranker scores a daily nine-strategy cross-section composed of eight delta-targeted short-put positions and a SKIP candidate" (Wysocki 2026, *Harvesting the VRP: A Learning-to-Rank Approach*, p. 1)

  - Its motivation is the desk's current situation: an always-on put book sizes the same whatever the premium.

    > "These passive harvesters have a structural limitation, since the position is held continuously regardless of the prevailing market regime, so capital is allocated identically on days when the per-trade premium is wide and on days when it is narrow" (Wysocki 2026, p. 2)

  - Its out-of-time Sharpe ratios of 4.3-5.8 come from 0DTE SPXW options and one hold-out year, so treat the size of the result with caution.

    > "Those out-of-time figures rest on a single calendar year." (Wysocki 2026, p. 43)

2.2 Volatility surface: term slope and skew

- The term-structure slope is the signal with the strongest evidence for index options.
  - **Outside the repo:** Johnson (2017, JFQA) finds that the VIX term-structure shape is about the price of variance risk, not about expected VIX changes. The slope predicts excess returns on S&P 500 variance swaps, VIX futures and straddles at all maturities, beyond other VRP proxies (abstract).
  - **Outside the repo:** Vasquez (2017) finds the same effect for single-stock straddles (abstract).
  - The library paper cites this literature.

    > "the slope of the term structure has been reported to predict the cross-section of subsequent option returns" (Wysocki 2026, p. 23)

- Skew is informative, but it is mostly the same risk as variance.
  - **Outside the repo:** Kozhan, Neuberger & Schneider (2013, RFS) find that the skew premium is over 40% of the slope of the S&P 500 implied-vol curve. Strategies that capture skew risk and hedge out variance risk "earn an insignificant risk premium" (abstract).
  - The PRISM whitepaper uses skew as an early regime warning. It is a self-published industry document whose results come from its own simulation backtest, so its claims are not independently checked.

    > "SPX skew steepens in the 2–3 weeks before elevated-volatility episodes" (Verma 2026, *PRISM*, p. 28)

- In the ranker, surface features matter most for choosing *which* put to sell. The RV-versus-IV gaps (the desk's VRP) are among the largest contributors.

  > "they are one of the largest contributors to model performance in the ablation panel of Section 6.3." (Wysocki 2026, p. 23)

2.3 Volatility of volatility (VVIX)

- **Outside the repo:** Park (2015, J. Financial Markets) finds that a rise in VVIX raises the prices of S&P 500 puts and VIX calls and "lowers their subsequent returns over the next three to four weeks". The result is robust to VRP, skew, jump risk and liquidity (abstract). For a put seller, high VVIX is a reason to sell more, not less.
- The library supports vol of vol as a separate state variable, not just a stand-in for the vol level.

> "there is a growing literature in finance that documents the emergence of volatility of volatility as a separate risk factor and its weak correlation with volatility risk" (Chong & Todorov 2024, *Volatility of Volatility and Leverage Effect from Options*, p. 1)

2.4 Calendar: weekends, overnight and announcements

- **Weekends (outside the repo):** Jones & Shemesh (2018, JF) find that "option returns are significantly lower over nontrading periods, the vast majority of which are weekends". They attribute this to persistent mispricing from the "incorrect treatment of stock return variance during periods of market closure", not to risk (abstract).
- **Overnight (outside the repo):** Muravyev & Ni (2020, JFE) find that delta-hedged S&P 500 index option returns average -1% per day close-to-open and $+0.3\%$ open-to-close (abstract). Option sellers earn the premium overnight.
- **FOMC (outside the repo):**
  - VIX and VIX futures fall significantly on FOMC announcement days (Fernandez-Perez, Frijns & Tourani-Rad 2017, abstract; Krieger, Mauck & Chen 2012, abstract).
  - Equities drift up before FOMC decisions (Lucca & Moench 2015, abstract).
  - Options whose life spans a political event are more expensive (Kelly, Pástor & Veronesi 2016, abstract).
- **In the ranker,** the calendar group's out-of-time effect is the largest of any feature group, although its walk-forward effect is modest. FOMC days are left out of the candidate set entirely.

> "although calendar's out-of-time drop of -2.50 is the largest in the matrix" (Wysocki 2026, p. 35) "FOMC days are excluded upstream from the candidate universe." (Wysocki 2026, p. 23)

2.5 Positioning and flow

- **Dealer gamma (outside the repo):**
  - When dealers are short gamma, their hedging trades go with the price and create intraday momentum (Baltussen, Da, Lammers & Martens 2021, JFE, abstract).
  - Market-maker hedge rebalancing raises stock volatility and the chance of large moves (Ni, Pearson, Poteshman & White 2021, RFS, abstract).
  - Barbon & Buraschi's *Gamma Fragility* is a working paper with a related result (abstract).
  - **Counterpoint:** Dim, Eraker & Vilkov find that high 0DTE open-interest gamma "does not propagate past volatility" (seminar abstract).
- **Crash-insurance demand (outside the repo):** Chen, Joslin & Ni (2019, RFS) build a measure from trading in deep out-of-the-money S&P 500 puts. Its tightening goes with more expensive options and higher risk premia across assets (NBER abstract). Gârleanu, Pedersen & Poteshman (2009) give the theory: end-user demand explains the expensiveness and skew of index options (abstract).

- **Put-call ratios:** the ranker uses them as sentiment.

  > "The PCR series carry sentiment-driven information beyond what is contained in the implied-volatility surface itself" (Wysocki 2026, p. 23)

2.6 Trend, regimes and momentum

- The ranker found nothing in trend features.

  > "Trend signals are a natural hypothesis for short-volatility strategy selection, but the sample provides no evidence that they help within-day selection at this threshold." (Wysocki 2026, p. 35)

- The jump-model paper times equity exposure, not options.

  > "a regime-switching investment strategy aimed at mitigating downside risk by reducing market exposure during anticipated unfavorable market regimes" (Shu, Yu & Mulvey 2024, *Downside Risk Reduction Using Regime-Switching Signals*, p. 1)

  - The desk's earlier report found statistical regime models detect crashes about two weeks late.

- OPHR uses RV-percentile momentum and mean reversion as straddle-timing baselines, on crypto options.

  > "Takes a long volatility position when the percentile momentum is high (expecting continuation), and a short volatility position when it is low." (Chen, Cai, Qin & An 2025, *OPHR*, p. 27) "Evaluating our approach using cryptocurrency options data from 2021-2024" (Chen et al. 2025, p. 1)

2.7 News, text and LLM agents

- News helps volatility forecasts only alongside a strong history-based model.

  > "News-only forecasts contain useful predictive information but generally do not outperform strong volatility-history benchmarks." (Rahimikia, Zohren & Poon 2026, *Realised Volatility Forecasting via Financial Word Embedding*, p. 1) "combining stock-related news forecasts with a strong volatility-history benchmark lowers forecast losses for several specifications and increases realised utility" (Rahimikia et al. 2026, p. 1)

- LLM agents do best on judgment calls, with the calculations and execution done by fixed code.

  > "The agentic architecture becomes competitive with this fully compiled benchmark when the agent is reserved for tender judgment and supported by pre-computed calculations and compiled unwind execution." (Cheng, Granger, Shi & Strela 2026, *Agents Are Not Algorithms*, p. 1)

2.8 Trades beyond long or short gamma

- **Choosing the strike or expiry, or skipping.** The ranker turns the signal into a choice among options rather than a yes/no decision (Wysocki 2026, p. 1).
- **VRP as a delta signal (outside the repo).** Bollerslev, Tauchen & Zhou (2009, RFS) find that a high VRP predicts high index returns, most strongly at quarterly horizons (abstract). That bears on whether to delta-hedge the short put, not only whether to sell it.
- **Writing calls or puts with the delta hedged (outside the repo).** In covered calls, the short-vol part earned a Sharpe of nearly 1.0, while the equity-reversal part added risk with little return (Israelov & Nielsen 2015, FAJ, abstract).

- **Volatility premia on other assets (outside the repo).** Shorting volatility across 34 markets in 4 asset classes gave a Sharpe of about 1.0 with substantial tail risk. Costs cut gross returns by 47% on average (Fallon, Park & Yu 2015, FAJ, abstract).
- **VIX futures basis (outside the repo).** Shorting VIX futures in contango and buying in backwardation, hedged with S&P futures, was profitable after costs from 2006 to 2011 (Simon & Campasano 2014, paper). It is a futures trade, not SPY options.
- **0DTE (outside the repo).** Vilkov's replication repository carries an August 2026 correction: a cost-scaling error charged spreads at 1/100 of their true size, and after the fix "no strategy or basket retains a positive net Sharpe ratio" (repo).

3 Desk data check: the weekend effect in `always25`

- **Data:** the registered one-contract `always25` book (2026-10-17-always-short), 2014-06 to 2024-12, P&L from each 10:15 decision to the next. Only intervals that hold the put and end without a trade, roll or stop are counted.

Interval (10:15 to 10:15)	Intervals	Sum, 1x	Mean	t-stat
Friday to Monday (incl. long weekends)	525	+$2,491	+$4.74	2.9
Monday to Tuesday	468	+$3,170	+$6.77	3.9
Tuesday to Wednesday	516	+$28	+$0.05	0.0
Wednesday to Thursday	494	−$263	−$0.53	−0.3
Thursday to Friday	351	−$918	−$2.62	−1.2

- **The weekend gain holds in both halves.** The mean weekend interval made $5.3 against $0.5 for other days in 2014-2019, and $4.1 against $1.8 in 2020-2024. The 2x run gives the same picture.
- **The sized book shows it too.** Over all intervals, including trade days, `hedged` from 2026-10-22-stress-sized-puts made a mean of $104 per weekend interval against $22 on one-day intervals.
- **Caveats:**
  - Five buckets were looked at, after reading the papers.
  - The t-stats assume independent days.
  - Marks are 10:15 mids. Each interval contains one overnight, so overnight and intraday can't be separated.
  - Intervals that end in a trade (mostly Friday rolls and stops) lost $3.3k in total. That loss is not counted above.

4 How the papers connect

Paper	Builds on / agrees with	Differs from	Key contribution
Wysocki 2026 (repo)	Vasquez 2017; Johnson 2017; VRP literature	Static put-writing (PUT index)	Ranks which put to sell or skip; calendar and surface matter, trend doesn't
Johnson 2017 (web)	Bakshi & Kapadia 2003	Expectations hypothesis for VIX	VIX slope predicts index variance and straddle returns beyond VRP
Park 2015 (web)	Chong & Todorov (repo): vol of vol is a separate factor	VIX-level timing	High VVIX means lower later put returns
Jones & Shemesh 2018; Muravyev & Ni 2020 (web)	Each other: variance is mis-timed in prices	Risk-based explanations	Option sellers earn over weekends and overnight
Kozhan et al. 2013 (web)	Gârleanu et al. 2009	Skew as a separate premium	Skew and variance premia are one risk
Baltussen et al. 2021; Ni et al. 2021 (web)	Gârleanu et al. 2009	Dim, Eraker & Vilkov (0DTE gamma)	Dealer hedging moves prices and volatility
Shu et al. 2024; OPHR 2025 (repo)	HMM regime literature	Wysocki 2026 (trend adds nothing)	Regime and momentum timing, tested off SPX options

- **Two lines agree:** the term-structure line (Vasquez, Johnson, Wysocki) and the calendar line (Jones & Shemesh, Muravyev & Ni, our data). Both rest on large samples and peer review.
- **Skew is the main disagreement.** PRISM treats it as an early warning signal. Kozhan et al. show it adds little premium beyond variance.
- **Dealer gamma is contested.** The momentum evidence is strong for futures. The 0DTE counter-evidence says new gamma doesn't amplify volatility.
- **The ML papers agree on one thing.** Option-based features beat price-based ones: Wysocki's ablation, and Bali et al. (2023) on equity options (abstract).

5 Relevance to the LS Gamma desk

These are suggestions. Thresholds, sizes and structures are the PM's call.

Context from the desk's own results:

- Gates that skip shorts have failed every time: the VIX term gate (twice) and the VIX/VIX3M entry gate. The 2026-10-18 diagnostic found stress filters remove profit.
- The VRP level barely ranks P&L.
- The always-on put works, and its weakness is size and concentration.

So new signals look more useful for **sizing, timing and choosing the option** than for deciding whether to trade.

Candidate	Trade	Data the desk has	Prior desk result	Suggested first test
Weekend / overnight	Roll or add puts on Fridays; avoid Friday exits	Snapshots (Friday expiries 1-60 DTE)	Data check above	Card: Friday-only roll day vs the 7-DTE rule, on `always25`
VVIX level / change	Size short puts up when VVIX is high	Live yfinance `^VVIX`; history needs a VVIX series	None	Card: VVIX-scaled size on the stress-sized book
VIX / term slope	Size, or pick the expiry (sell where the curve is steepest)	VIX, VIX9D, VIX3M; snapshot term structure	Gates failed (used to skip)	Card: slope as a size multiplier, not a gate
Skew / strike choice	Pick the put delta each day (rank or rule)	Snapshot smile fits	25-delta beat ATM (costs)	Diagnostic: forward P&L by delta bucket against skew
VRP for delta	Hedge less when VRP is high	Existing signal	`unhedged` P&L was 75% delta	Card: hedge ratio conditional on VRP
FOMC / CPI days	Hold or add short vol into FOMC; skip entry on CPI	Needs an event calendar	None	Diagnostic on `always25` P&L by event day
Cross-asset VRP	Short puts on QQQ/IWM/GLD/ TLT	R2 trades for 5 more ETFs; needs snapshot builds	None	Build one ETF (QQQ or TLT) and rerun `always25`
Dealer gamma	Short gamma when dealers are long gamma	No history (R2 has no open interest); IBKR open interest live only	None	Start logging open interest now
News / LLM	Event judgment layer	None	None	Later; only as an input to rules

- **Cheapest first:** weekend timing, VVIX and slope sizing use data the desk already loads.
- **The FOMC check** needs only a list of dates.
- **Cross-asset** needs new snapshot builds but uses data already on R2.
- **Not recommended now:**
  - 0DTE: losing after costs, and needs intraday infrastructure;
  - skew as a standalone trade: redundant with VRP;
  - trend or HMM timing: no evidence for option selection, and too slow for crashes.

6 Gaps and open questions

- **Index-specific weekend evidence.** Jones & Shemesh's abstract doesn't confirm SPX coverage, and our check is one book with 10:15 marks.
- **Whether a weekend trade survives costs.** A weekend-only trade would pay the spread every week, and $4.74 a weekend on one contract is below a round-trip spread plus $1.30 commission. The edge is more likely in when to roll and how much to hold.
- **Sign and horizon of the slope signal for SPY puts.** Johnson's evidence is for variance swaps, VIX futures and straddles, not 25-delta puts held to 7 DTE.
- **Missing histories.**
  - VVIX history isn't in R2's CBOE file (VIX, VXO, VXN, VXD only).

  - Open interest isn't anywhere in R2.
  - R2 has no trade direction (no quotes), so a Chen-Joslin-Ni demand measure can't be built cleanly.

- **Margin and fills.** Cross-asset and size-up signals add margin, which the backtest doesn't model.
- **One sample for every check.** Every new signal is another trial on the same 2014-2024 window (76 trials registered). The 2025-2026 hold-out can be used once per card.

7 Sources

Repo papers used

- `Harvesting-the-Volatility-Risk-Premium-A-Learning-to-Rank-Approach.pdf`: Wysocki 2026, pp. 1, 2, 23, 35, 43
- `Volatility-of-Volatility-and-Leverage-Effect-from-Options.pdf`: Chong & Todorov 2024, p. 1
- `OPHR-Mastering-Volatility-Trading-with-Multi-Agent-Deep-Reinforcement-Learning.pdf`: Chen, Cai, Qin & An 2025, pp. 1, 27
- `Downside-Risk-Reduction-Using-Regime-Switching-Signals-A-Statistical-Jump-Model-Approach.pdf`: Shu, Yu & Mulvey 2024, p. 1
- `Realised-Volatility-Forecasting-Machine-Learning-via-Financial-Word-Embedding.pdf`: Rahimikia, Zohren & Poon 2026, p. 1
- `Agents-Are-not-Algorithms.pdf`: Cheng, Granger, Shi & Strela 2026, p. 1
- `PRISM-A-Probabilistic-Regime-Integrated-Scalping-Model-for-Derivatives-Arbitrage.pdf`: Verma 2026, p. 28 (industry whitepaper)

Repo papers checked with nothing new on trade signals

- **Hedging** (covered in earlier reports): the Deep Hedging papers (`Deep-Hedging.pdf`, `Deep-Bellman-Hedging.pdf`, `Learning-to-Trade-II-Deep-Hedging.pdf`, `Deep-Hedging-Learning-to-Simulate-Equity-Option-Markets.pdf`, `Deep-Hedging-of-Derivatives-Using-Reinforcement-Learning.pdf`, `Deep-Hedging-with-Options-Using-the-Implied-Volatility-Surface.pdf`, `Enhancing-Deep-Hedging-of-Options-with-Implied-Volatility-Surface-Feedback-Information.pdf`, `Deep-Hedging-with-Reinforcement-Learning-A-Practical-Framework-for-Option-Risk-Management.pdf`, `Gamma-and-Vega-Hedging-Using-Deep-Distributional-Reinforcement-Learning.pdf`, `Application-of-Deep-Reinforcement-Learning-to-At-the-Money-S&P-500-Options-Hedging.pdf`, `Is-the-Difference-between-Deep-Hedging-and-Delta-Hedging-a-Statistical-Arbitrage.pdf`), `Hedging-with-Linear-Regressions-and-Neural-Networks.pdf` and `When-You-Hedge-Discretely-...pdf`.
- **RV forecasting** (covered in earlier reports): `A-machine-learning-approach-to-volatility-forecasting.pdf`, `Foundation-Time-Series-AI-Model-for-Realized-Volatility-Forecasting.pdf`, `Improving-S&P-500-Volatility-Forecasting-through-Regime-Switching-Methods.pdf`, `Novel-Optimization-Approach-...-Deep-Reinforcement-Learning-Model.pdf`, `Realized-GARCH,-CBOE-VIX,-and-the-Volatility-Risk-Premium.pdf`, `Volatility-Forecasting-and-Volatility-Risk-Premium.pdf`, `Time-Series-Embedding-and-Combination-of-Forecasts-...pdf` and `Reinforcement-Learning-based-Dynamic-Model-Selection-...pdf`.
- **Regimes and allocation:** `Clustering-Market-Regimes-Using-the-Wasserstein-Distance.pdf`, `Learning-Hidden-Markov-Models-with-Persistent-States-by-Penalizing-Jumps.pdf` and `Dynamic-Investment-Strategies-Through-Market-Classification-and-Volatility-...pdf`.

- **Pricing and identification:** `Identifying-the-Volatility-Risk-Price-Through-the-Leverage-Effect.pdf`.
- **Backtest method** (covered in earlier reports): `A-Novel-Approach-to-Trading-Strategy-Parameter-Optimization-...pdf` and `The-GT-Score-...pdf`.
- **Survey:** `From-Deep-Learning-to-LLMs-A-Survey-of-AI-in-Quantitative-Investment.pdf` has sentiment and news methods for stock prediction but no option-trading evidence.

Web sources (outside the repo)
Calendar and timing

- Jones & Shemesh 2018, JF
- Muravyev & Ni 2020, JFE
- Lucca & Moench 2015, JF
- Kelly, Pástor & Veronesi 2016, JF
- Fernandez-Perez, Frijns & Tourani-Rad 2017, JEF
- Krieger, Mauck & Chen 2012, FMPM

Surface, VVIX and VRP

- Johnson 2017, JFQA
- Vasquez 2017, JFQA
- Kozhan, Neuberger & Schneider 2013, RFS
- Park 2015, JFM
- Bollerslev, Tauchen & Zhou 2009, RFS
- Bakshi & Kapadia 2003, RFS

Positioning and flow

- Baltussen, Da, Lammers & Martens 2021, JFE
- Ni, Pearson, Poteshman & White 2021, RFS
- Barbon & Buraschi, Gamma Fragility (working paper)
- Dim, Eraker & Vilkov, 0DTEs (seminar abstract)
- Chen, Joslin & Ni 2019, RFS (NBER)
- Gârleanu, Pedersen & Poteshman 2009, RFS

Option writing, cross-asset and 0DTE

- Bali, Beckmeyer, Mörke & Weigert 2023, RFS
- Israelov & Nielsen 2015, FAJ
- Fallon, Park & Yu 2015, FAJ
- Simon & Campasano, VIX futures basis
- Vilkov, 0DTE trading rules (replication and correction)

Suggested papers to add to `docs/papers/`

1. **Johnson 2017,** *Risk Premia and the VIX Term Structure*: the cleanest index-level signal beyond VRP.
2. **Jones & Shemesh 2018,** *Option Mispricing around Nontrading Periods*: the weekend effect our data shows.
3. **Muravyev & Ni 2020,** *Why Do Option Returns Change Sign from Day to Night?*: when the SPX premium is earned.
4. **Park 2015,** *Volatility-of-Volatility and Tail Risk Hedging Returns*: VVIX, already in the live pipeline.
5. **Kozhan, Neuberger & Schneider 2013,** *The Skew Risk Premium in the Equity Index Market*: why skew isn't a separate edge.